

# Linear regression

[MI](#)

**Linear Regression (LR)** is a type of supervised machine learning algorithm that predicts outcomes based on previous data by finding the best-fit line between input and output variables.

## Types of Linear Regression

| Type                     | Shape of Relationship      | Example               | Equation                          |
|--------------------------|----------------------------|-----------------------|-----------------------------------|
| Simple Linear            | Straight line              | Hours studied → Marks | $Y = mX + c$                      |
| Multiple Linear          | Flat plane (many features) | Area, rooms → Price   | $Y = m_1X_1 + m_2X_2 + c$         |
| <b>Polynomial Linear</b> | <b>Curved line</b>         | Time → Plant growth   | $Y = b_0 + b_1X + b_2X^2 + \dots$ |

## Assumptions of linear regression

Linear regression is built on several key assumptions that make its estimates valid and reliable.

The main assumptions are that the model is:

### 1. Linearity

The model assumes a straight-line relationship between each independent variable and the dependent variable.

#### **If violated:**

- Coefficients become **biased**.
- Model gives **poor predictions** (underfitting).
- **Hypothesis tests and confidence intervals** may become invalid.

#### **How to Check Linearity**

##### **- Scatter plots:**

- Plot Y vs each X. A straight-line pattern means linearity; curved shapes mean violation.

##### **- Residual plots:**

- Plot residuals vs predicted values or vs each X. Random scatter around zero = good. Curves or patterns = linearity violated.

##### **- Polynomial terms:**

- Add squared or higher-order terms and compare model fit. If the polynomial model fits

much better, the original linearity assumption was likely violated.

### **What to Do When Linearity Fails**

#### **- Transformations:**

- Use log, square-root, or inverse transformations on Y or X to make the relationship more linear.

#### **- Polynomial regression:**

- Add squared or higher-order terms to capture curved relationships.

#### **- Piecewise regression:**

- Break X into intervals and fit separate linear lines for each segment.

#### **- Non-parametric methods:**

- Use flexible models like GAMs, splines, or kernel regression that don't require strict linearity.

## **2. Normality of residuals**

Residuals should be normally distributed with mean 0 and constant variance.

### **If violated:**

- **Hypothesis tests become unreliable** → t-tests & F-tests may give wrong p-values.

- **Confidence intervals become inaccurate** → too wide or too narrow.

- **Model performance weakens** → may indicate poor fit and lower prediction quality.

### **How to Check Normality of Residuals : ==**

#### **- Histogram:**

- Plot a histogram of residuals. A bell-shaped pattern suggests normality.

#### **- Q-Q plot:**

- If residual points lie roughly on a straight line, normality is likely; strong bends mean non-normality.

#### **- Statistical tests:**

- Use tests like Omnibus, Jarque-Bera, or Shapiro-Wilk to formally check if residuals differ from a normal distribution.

### **==What to Do When Normality of Residuals Fails :**

#### **- Model selection:**

- Use cross-validation, AIC, or BIC to choose a model that better handles non-normal residuals.

#### **- Robust regression:**

- Use methods like M-estimation, LMS, or LTS, or apply variable transformations to reduce sensitivity to non-normality.

#### **- Non-parametric methods:**

- Use flexible models such as GAMs, splines, or kernel regression that do not assume normal residuals.

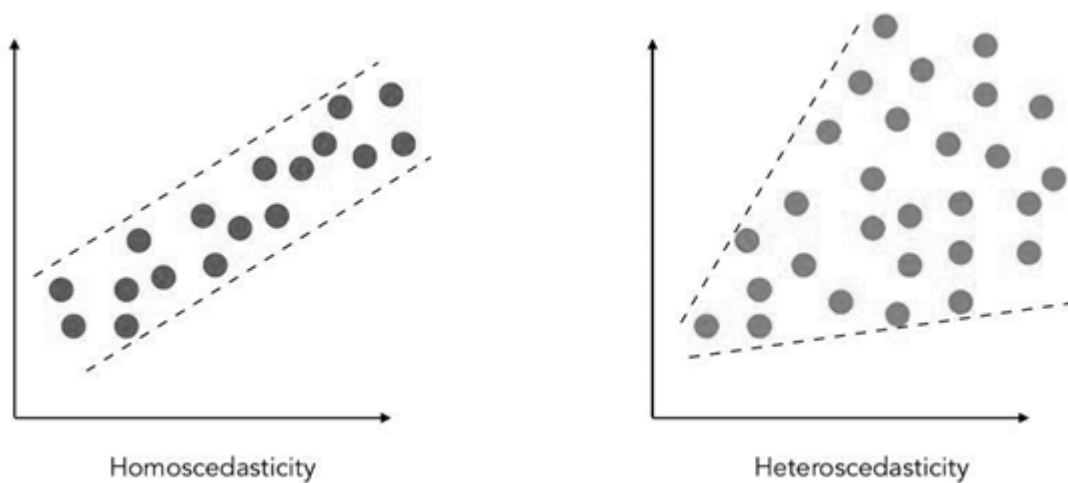
#### **- Bootstrapping:**

- Use bootstrap-based confidence intervals and tests, which work even when residuals are not normal.

**==Note: ==**

**Normality is less important for large samples due to the Central Limit Theorem.**

### 3. **Homoscedastic** (residuals have constant variance)



Residuals should have **constant variance** across all values of the independent variables. If the spread of residuals increases or decreases, it indicates **heteroscedasticity**.

#### **If violated:**

- **Inefficient estimates:** Coefficients stay unbiased, but standard errors become inflated → lower test power.
- **Inaccurate hypothesis tests:** t-tests and F-tests may give wrong p-values.
- **Invalid confidence intervals:** Intervals may be too wide or too narrow, harming interpretation.

### 4. No autocorrelation

No autocorrelation means:

- **Error terms are independent of each other**
- **Residuals are random (no pattern, no trend)**
- Required for classical linear regression assumptions

Example:

If today's error has no relation with tomorrow's error → **no autocorrelation**.

#### **1. If No-Autocorrelation Assumption Is Violated**

- Standard errors become wrong → **t-test & p-value unreliable**
- Coefficients become **inefficient**
- Model may show **false significance**
- Forecasting quality drops
- Shows **time pattern or missing variables** in the model

#### **2. How to Check Autocorrelation**

- **Durbin-Watson test** →  $\sim 2$  = no autocorrelation
- **Residuals vs Time plot** → random = no autocorrelation
- **ACF plot** → big spikes = autocorrelation
- **Ljung-Box test** → detects autocorrelation at many lags

#### **3. What to Do If Autocorrelation Exists**

- Add **trend, seasonality, lag variables**

- Use **ARIMA / ARIMAX / GLS / Cochrane–Orcutt**
- Transform data: **first difference, log**
- Use **Newey–West robust errors**

## 5. **Non-multicollinear** (predictors are not highly correlated)

### **Multicollinearity**

Multicollinearity occurs when **two or more independent variables in a regression model are highly correlated**, making it difficult to separate and measure their individual effects on the dependent variable.

#### **\*\*When is Multicollinearity Bad?**

Multicollinearity is bad when your goal is *Inference*

Inference focuses on understanding the relationships between the variables in a model. It aims to draw conclusions about the underlying population or process that generated the data.

Because inference focuses on **understanding relationships**, multicollinearity becomes a problem since it:

- Makes coefficient estimates **unstable**
- Inflates **standard errors**
- Makes **p-values unreliable**
- Makes it hard to know **which variable actually matters**
- Reduces interpretability → can't understand individual variable effects

👉 **So, for inference, multicollinearity is a serious problem.**

## **When is Multicollinearity NOT a big problem?**

### **When your goal is *Prediction***

Prediction cares about **accuracy**, not coefficient interpretation.

Prediction focuses on using a model to make accurate forecasts or estimates for new, unseen data. It aims to generalize the model to new instances, based on the patterns observed in the training data.

Even if predictors are correlated:

- Model can still predict well
- Coefficients don't need to be interpretable
- Multicollinearity does *not* affect prediction error much

👉 **So for prediction tasks, multicollinearity is usually acceptable.**

## **FINAL SUMMARY**

- **Inference** → **MC is BAD** because you need to interpret coefficients.
- **Prediction** → **MC is OK** because you only need accurate forecasts.

### **Effects of Multicollinearity**

#### **1. Hard to identify important predictors**

When predictors are highly correlated, it becomes difficult to know which variable

actually influences the dependent variable, because they carry overlapping information.

## 2 . Inflated standard errors:

Multicollinearity can lead to larger standard errors for the regression coefficients, which decreases the statistical power and can make it challenging to determine the true relationship between the independent and dependent variables.

## 3. Unstable and unreliable estimates:

The regression coefficients become sensitive to small changes in the data, making it difficult to interpret the results accurately.

### Perfect Multicollinearity (Short Definition)

Perfect multicollinearity occurs when **one independent variable is an exact linear combination of one or more other independent variables**.

Because the relationship is *perfect*, the model cannot separate their effects, making the regression coefficients **impossible to estimate**.

## Note

- **Perfect multicollinearity** → OLS fails completely
- **Imperfect (high) multicollinearity** → OLS works, but estimates are **unstable and unreliable**

### Ways to detect Multicollinearity

- code --> [Code](#)
  - **Correlation matrix**
    - Correlation helps detect multicollinearity by showing how strongly predictors are related. High pairwise correlations ( $\approx 0.8-0.9+$ ) suggest multicollinearity, which makes coefficients unstable. However, correlation is not enough—**VIF** and **condition number** give a better assessment.
  - **Variance Inflation Factor (VIF)**
    - **Variance Inflation Factor (VIF)** measures how much **multicollinearity inflates the variance** of a regression coefficient.(-)
    - **VIF  $\approx 1$**  → no multicollinearity
    - **VIF  $> 5$  (or 10)** → serious multicollinearity, unstable coefficients
  - **Condition number**
    - **Condition Number** measures how sensitive a regression model is to small changes in data and is used to detect **multicollinearity**.(-)
    - **Large condition number** → numerical instability and unstable coefficients
      - **$> 30$**  → serious multicollinearity (rule of thumb)

### Ways to reduce or handle Multicollinearity

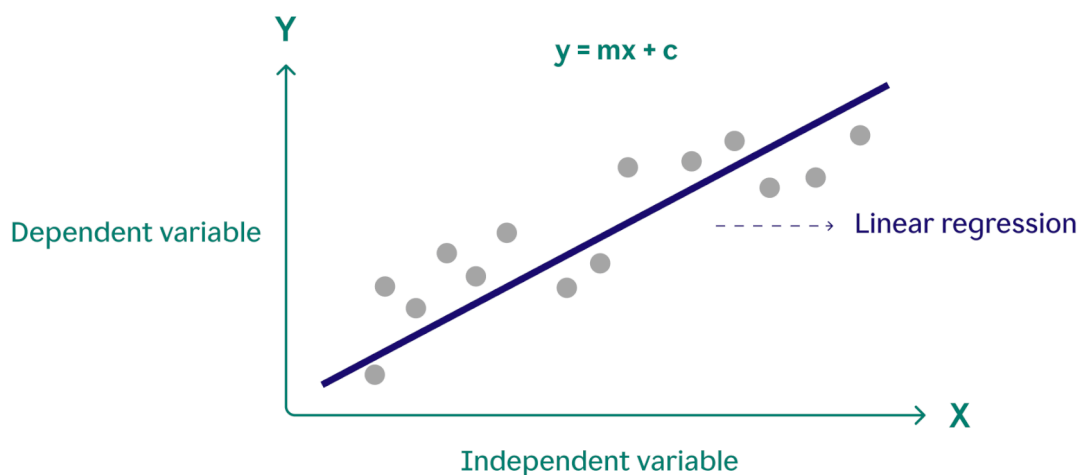
- Collect more data
- Remove one of the highly correlated variables

- Combine correlated variables
- Use Partial Least Squares (PLS) regression

## Simple Linear Regression

Simple Linear Regression is a **supervised machine learning algorithm** that shows the **relationship between one independent variable (X) and one dependent variable (Y)** by fitting a **straight line** through the data.

It is used to **predict a continuous output** based on a single input feature.



Equation:

$$Y = mX + b$$

where:

- **Y** → Predicted output
- **X** → Input variable
- **m** → Slope of the line (shows how much Y changes when X changes)  
In simple words --> **m** defines how strongly or how much the output (Y) changes when the input (X) changes.
- **b** → Intercept (value of Y when X = 0) it's **something that remains even if  $m \times X = 0$**

## How to find m and b ?

There are mainly two common methods\*\* to find them

1. Closed-form (OLS)

## 2. Gradient descent

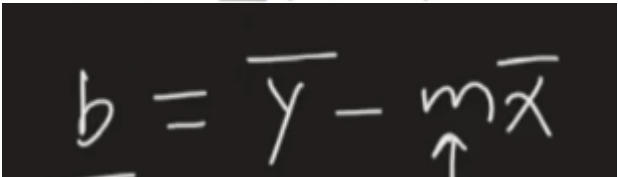
| Method               | Description                      | Used When                            |
|----------------------|----------------------------------|--------------------------------------|
| Least Squares Method | Uses direct mathematical formula | Small or medium datasets             |
| Gradient Descent     | Iteratively improves m and c     | Large datasets / used in ML training |

### Closed-form (OLS)

OLS finds the line where the total distance between the real data points and the predicted line (vertically measured) is as **small as possible**.

It's called "**least squares**" because it **minimizes the sum of squared errors** — this avoids negative and positive errors canceling each other out.

Formula :

$$m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$


A handwritten formula on a black background:  $b = \bar{y} - m\bar{x}$ . An upward-pointing arrow is drawn under the  $m$  term.



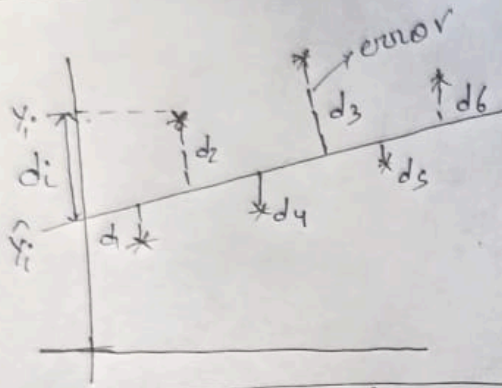
OLS

To find  $m$  &  $b$

$$b = \bar{y} - m\bar{x}$$

$$m = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Let,



$$E = d_1 + d_2 + d_3 + \dots + d_n$$

$$E = d_1^2 + d_2^2 + d_3^2 + \dots + d_n^2$$

$$E = \sum_{i=1}^n d_i^2 \quad \leftarrow \text{error function}$$

Why square, why not mod

$$\text{means } E = |d_1| + |d_2| + |d_3|$$

~~because it not diff<sup>n</sup>~~

from eq,

$$E = \sum_{i=1}^n d_i^2$$

$$d_i = (y_i - \hat{y}_i)$$

$$\text{So, } E = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

as we know

$$\hat{y}_i = mx_i + b$$

So,

$$E(m, b) = \sum_{i=1}^n (y_i - mx_i - b)^2$$

So, we know we have find a perfect value of  $m$  &  $b$  to reduce error which also known as  $J(\theta_0, \theta_1)$

Now,

we have to find out the partial derivative of  $\frac{\partial E}{\partial m}$ ,  $\frac{\partial E}{\partial b}$



So,

$$\frac{\partial E}{\partial b} = \frac{\partial}{\partial b} \sum_{i=1}^n (y_i - mx_i - b)^2 = 0$$

$$= \sum \frac{\partial}{\partial b} (y_i - mx_i - b)^2 = 0$$

$$= \sum -2 (y_i - mx_i - b) = 0$$

divid by -2 both side.

$$= \sum (y_i - mx_i - b) = 0$$

$$= \sum y_i - \sum mx_i - \sum b = 0$$

now divide by  $n$ , where  $n$  is the total number of rows/student

$$\frac{\sum_{i=1}^n y_i}{n} - \frac{\sum mx_i}{n} - \frac{\sum b}{n} = \frac{0}{n}$$

$$\frac{\sum b}{n} = \frac{n \cdot b}{n}$$

$$= \bar{y} - m\bar{x} = b$$

$$b = \bar{y} - m\bar{x}$$

hence proved / define

Now, we will use the equation of  $b$  in the equation  $E$

$$E = \sum (y_i - mx_i - \bar{y} + m\bar{x})^2$$

Now, we have to find out partial derivative

$$\frac{\partial E}{\partial m} = \sum \frac{\partial}{\partial m} (y_i - mx_i - \bar{y} + m\bar{x})^2 = 0$$

$$= \sum -2 (y_i - mx_i - \bar{y} + m\bar{x}) (x_i - \bar{x}) = 0$$

divide both side with  $-2$

$$= \sum (y_i - mx_i - \bar{y} + m\bar{x}) (x_i - \bar{x}) = 0$$

$$= \sum [(y_i - \bar{y}) - m(x_i - \bar{x})] (x_i - \bar{x}) = 0$$

$$= \sum [(y_i - \bar{y}) (x_i - \bar{x}) - m(x_i - \bar{x})^2] = 0$$

$$= \sum (y_i - \bar{y}) (x_i - \bar{x}) = m \sum (x_i - \bar{x})^2$$

$$m = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Hence, define.

Code!.

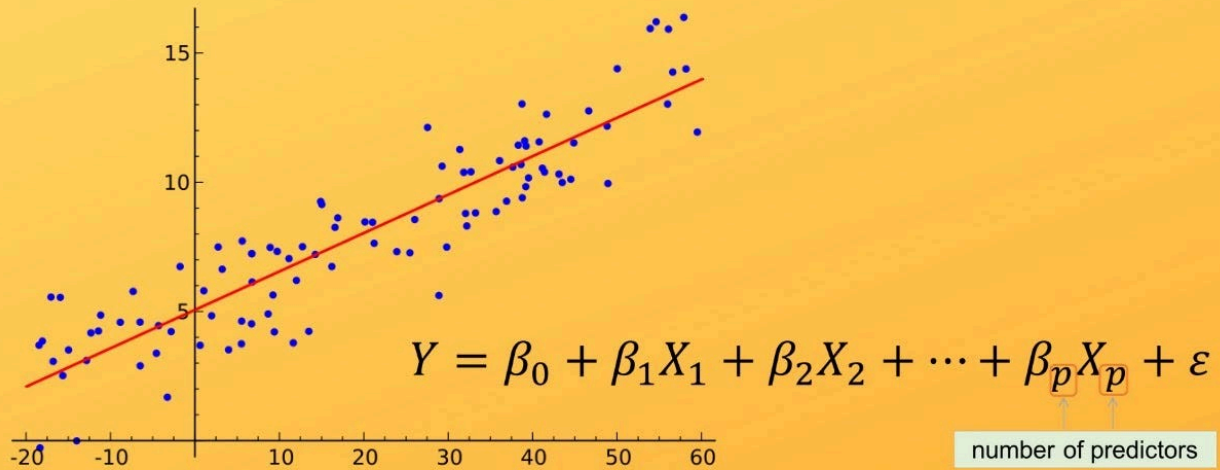
Making a class

## MLR (Multiple Linear Regression)

**Multiple Linear Regression** is a supervised learning technique used to predict a **continuous numerical value** using **two or more input features**.

It is an extension of **simple linear regression**, which uses only one feature.

# Multiple Linear Regression



Using **OLS**



## OLS for Multiple LR

As we know,  $\rightarrow \hat{Y}_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_m X_{im}$

for the first row to predict the outcome,  
So, similarly.

$$\hat{Y}_1 = \beta_0 + \beta_1 X_{11} + \beta_2 X_{12} + \beta_3 X_{13} + \dots + \beta_m X_{1m}$$

$$\hat{Y}_2 = \beta_0 + \beta_1 X_{21} + \beta_2 X_{22} + \beta_3 X_{23} + \dots + \beta_m X_{2m}$$

$$\hat{Y}_3 = \beta_0 + \beta_1 X_{31} + \beta_2 X_{32} + \beta_3 X_{33} + \dots + \beta_m X_{3m}$$

$$\vdots$$
$$\hat{Y}_n = \beta_0 + \beta_1 X_{n1} + \beta_2 X_{n2} + \beta_3 X_{n3} + \dots + \beta_m X_{nm}$$

So,

$$\hat{Y} = \begin{bmatrix} \hat{Y}_1 \\ \hat{Y}_2 \\ \vdots \\ \hat{Y}_n \end{bmatrix}_{(n \times 1)} = \begin{bmatrix} \beta_0 + \beta_1 X_{11} + \beta_2 X_{12} + \dots + \beta_m X_{1m} \\ \beta_0 + \beta_1 X_{21} + \beta_2 X_{22} + \dots + \beta_m X_{2m} \\ \vdots \\ \beta_0 + \beta_1 X_{n1} + \beta_2 X_{n2} + \dots + \beta_m X_{nm} \end{bmatrix}_{(n \times 1)}$$

$$\begin{bmatrix} 1 & X_{11} & X_{12} & X_{13} & \dots & X_{1m} \\ 1 & X_{21} & X_{22} & X_{23} & \dots & X_{2m} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & X_{n2} & X_{n3} & \dots & X_{nm} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_m \end{bmatrix}$$

$\downarrow$   $X$   $\downarrow$   $\beta$

$$\hat{Y} = X\beta$$

$\nwarrow$   $\searrow$   
 $n \times (m+1)$   $(m+1) \times 1$

$\downarrow$   $\nwarrow$   $\searrow$   
 $[n \times 1]$  size Eq (1)

As we know the Error

$$E = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

we have to min<sup>n</sup> the E but here we have to form the matrix

$$\text{Matrix } Y(\text{actual}) = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{bmatrix}, \text{ Matrix } \hat{Y}(\text{pred}) = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix}$$

$$\text{Matrix } e = Y - \hat{Y} = \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix}$$

Now, we Transpose e and again multiply with e

$$e^T e = \begin{bmatrix} y_1 - \hat{y}_1 & y_2 - \hat{y}_2 & y_3 - \hat{y}_3 & \dots & y_n - \hat{y}_n \end{bmatrix}_{(1 \times n)} \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ y_3 - \hat{y}_3 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix}_{(n \times 1)}$$

$$e^T e = (y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 + \dots + (y_n - \hat{y}_n)^2$$

$$e^T e = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

So,

$$E = e^T e$$

we have to min<sup>n</sup> these.  
eq (2)



So we re-write the equation

$$E = e^T e$$

$$E = (y - \hat{y})^T (y - \hat{y})$$

$$E = (Y - \hat{Y})^T (Y - \hat{Y}) \Rightarrow (Y^T - \hat{Y}^T) (Y - \hat{Y})$$

$$= Y^T Y - Y^T \hat{Y} - \hat{Y}^T Y + \hat{Y}^T \hat{Y}$$

$$E = Y^T Y - 2Y^T \hat{Y} + \hat{Y}^T \hat{Y}$$

eq (3)

Now,

Apply  $X\beta$  at place of  $\hat{Y}$  in eq 3

$$E = Y^T Y - 2Y^T X\beta + (X\beta)^T (X\beta)$$

$$E = Y^T Y - 2Y^T X\beta + \beta^T X^T X\beta$$

eq (4)

$$E(\beta)$$

→ We have to find such value for  $\beta$  so the  $E$  will min<sup>m</sup>

→ for that we have to  $\frac{dE}{d\beta} = 0$

$$\text{So, } \frac{dE}{d\beta} = 0 - 2Y^T X\beta + 2\beta^T X^T X = 0$$

$$\begin{aligned} \frac{d}{d\beta} \beta^T X^T X \beta &\downarrow \\ X^T A X &= 2X^T A \\ \text{Similarly} &= 2\beta^T X^T X \end{aligned}$$

$$\begin{aligned} \cancel{Y^T \hat{Y}} &= \cancel{\hat{Y}^T Y} \\ Y^T \hat{Y} &= \hat{Y}^T Y \\ A^T B &= B^T A \\ Y^T X \beta &= Y^T X \beta \\ (1 \times n) \quad n(m+1) \quad (m+1) \times 1 \\ &\quad \downarrow \quad \downarrow \\ &\quad 1 \times n \quad [n \times 1] \\ &\quad \text{[ ] scalar} \\ &\quad \text{So, } T \text{ is always} \\ &\quad \text{similar} \end{aligned}$$



$$2\beta^T X^T X = 2Y^T X$$

$$\beta^T X^T X = Y^T X$$

Now mul<sup>n</sup> by  $(X^T X)^{-1}$  both side.

$$\beta^T X^T X (X^T X)^{-1} = Y^T X (X^T X)^{-1}$$

$$\beta^T I = Y^T X (X^T X)^{-1}$$

$$\beta^T = Y^T X (X^T X)^{-1}$$

Applying both side transpose.

$$(\beta^T)^T = \left[ \underbrace{Y^T X}_A \underbrace{(X^T X)^{-1}}_B \right]^T$$

$$\beta = \left[ (X^T X)^{-1} \right]^T (Y^T X)^T$$

$$\beta = \left[ (X^T X)^{-1} \right]^T X^T Y$$

$$\beta = (X^T X)^{-1} X^T Y \quad \text{final eq.}$$

$(X^T X)^{-1}$  is symmetric

$$\left[ (X^T X)^{-1} \right]^T = (X^T X)^{-1}$$

Let,

$$X^T X = A$$

$$A^T = A$$

$$A A^{-1} = I$$

$$(A A^{-1})^T = I^T$$

$$(A^{-1})^T A^T = I$$

$$(A^{-1})^T A = I$$

$$(A^{-1})^T A A^{-1} = I A^{-1}$$

$$(A^{-1})^T I = A^{-1}$$

$$(A^{-1})^T = A^{-1}$$

So,

$$\left[ (X^T X)^{-1} \right]^T = (X^T X)^{-1}$$